National Technical University of Athens

School of Electrical and Computer Engineering

Doctor of Philosophy Program

Postgraduate Course:

228: Game Theory

Semester Project

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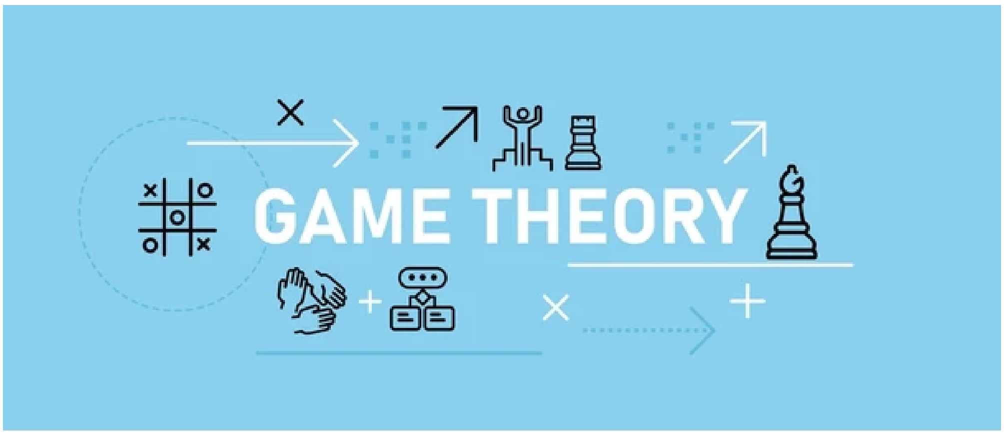

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Distributed Nash Consensus Control of Multi-Agent Systems Using an LQR-Based Approach

Agent Dynamics

Consider a network of N interacting agents described by the linear dynamics

$$\dot{x}_i = Ax_i + Bu_i + w_i, \qquad i = 1, \ldots, N, \tag{1}$$

where

$$x_i \in \mathbb{R}^n$$

denotes the state vector,

$$u_i \in \mathbb{R}^m$$

denotes the control input, and

$$w_i$$

represents bounded disturbances.
The communication topology is modeled by a connected graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E})$$

with Laplacian matrix

$$L.$$

Consensus Error

Define the local disagreement state of agent i as

$$e_i = \sum_{j \in \mathcal{N}_i} (x_i - x_j). \tag{2}$$

The vector e_i quantifies the deviation of agent i from its neighboring agents.
Stacking all states into

$$x = \begin{bmatrix} x_1^T & x_2^T & \cdots & x_N^T \end{bmatrix}^T, \tag{3}$$

the global consensus error becomes

$$e = (L \otimes I_n)x. \tag{4}$$

Consensus is achieved whenever

$$e = 0, \tag{5}$$

which implies

$$x_1 = x_2 = \cdots = x_N. \tag{6}$$

Local Nash Cost Function

Each agent seeks to minimize the infinite-horizon quadratic performance index

$$J_i = \int_0^\infty \left(e_i^T Q e_i + u_i^T R u_i \right) dt, \tag{7}$$

where

$$Q = Q^T > 0, \qquad R = R^T > 0.$$

The first term penalizes disagreement among neighboring agents, whereas the second term penalizes excessive control effort.

Since e_i depends on neighboring states, the cost function of each player depends on the actions of all neighboring agents.

Consequently,

$$J_i = J_i(u_i, u_{-i}), \tag{8}$$

which naturally defines a dynamic non-cooperative game.

Global Dynamic System

Define the aggregate state vector

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}. \tag{9}$$

The global dynamics can be written compactly as

$$\dot{x} = \mathcal{A}x + \mathcal{B}u, \tag{10}$$

where

$$\mathcal{A} = I_N \otimes A, \tag{11}$$

and

$$\mathcal{B} = I_N \otimes B. \tag{12}$$

Global Performance Index

Summing the local cost functions yields

$$J = \int_0^\infty \left(x^T (L \otimes I_n)^T Q (L \otimes I_n) x + u^T R u \right) dt. \tag{13}$$

Define the aggregate weighting matrix

$$Q_N = (L \otimes I_n)^T Q (L \otimes I_n). \tag{14}$$

Then the performance index becomes

$$J = \int_0^\infty \left(x^T Q_N x + u^T R u \right) dt. \tag{15}$$

This formulation transforms the consensus problem into a structured LQR optimization problem.

Hamilton–Jacobi–Bellman Equation

Let

$$V(x) \tag{16}$$

denote the value function.

The Hamiltonian associated with the game is

$$H = x^T Q_N x + u^T R u + \nabla V^T (\mathcal{A} x + \mathcal{B} u). \tag{17}$$

The HJB equation is

$$0 = \min_u \left\{ x^T Q_N x + u^T R u + \nabla V^T (\mathcal{A} x + \mathcal{B} u) \right\}. \tag{18}$$

Optimal Nash Strategy

Differentiating the Hamiltonian with respect to the control input gives

$$\frac{\partial H}{\partial u} = 2Ru + \mathcal{B}^T \nabla V. \tag{19}$$

The first-order optimality condition is

$$\frac{\partial H}{\partial u} = 0. \tag{20}$$

Hence,

$$2Ru + \mathcal{B}^T \nabla V = 0. \tag{21}$$

Solving for the optimal control yields

$$u^\star = -\frac{1}{2} R^{-1} \mathcal{B}^T \nabla V. \tag{22}$$

Quadratic Value Function

Assume a quadratic value function

$$V(x) = x^T P x, \tag{23}$$

where

$$P = P^T > 0.$$

Its gradient is

$$\nabla V = 2Px. \tag{24}$$

Substituting into the optimal control law gives

$$u^\star = -R^{-1}\mathcal{B}^T P x. \tag{25}$$

Define

$$K = R^{-1}\mathcal{B}^T P. \tag{26}$$

Therefore,

$$u^\star = -Kx. \tag{27}$$

Algebraic Riccati Equation

Substituting the quadratic value function into the HJB equation yields the Algebraic Riccati Equation

$$\mathcal{A}^T P + P\mathcal{A} - P\mathcal{B}R^{-1}\mathcal{B}^T P + Q_N = 0. \tag{28}$$

The unique positive-definite solution

$$P > 0$$

determines the optimal Nash gain

$$K = R^{-1}\mathcal{B}^T P. \tag{29}$$

Distributed Nash Consensus Controller

The centralized feedback law can be rewritten in distributed form using the consensus error

$$u_i^\star = -Ke_i. \tag{30}$$

Substituting the definition of e_i yields

$$u_i^\star = -K \sum_{j \in \mathcal{N}_i} (x_i - x_j). \tag{31}$$

This controller depends only on local neighboring information and is therefore fully distributed.

Closed-Loop Dynamics

Substituting the distributed Nash controller into the system dynamics gives

$$\dot{x}_i = Ax_i - BK \sum_{j \in \mathcal{N}_i} (x_i - x_j) + w_i. \tag{32}$$

The corresponding global dynamics become

$$\dot{x} = \Big(I_N \otimes A - L \otimes BK\Big)x + w. \tag{33}$$

Lyapunov Stability Analysis

Consider the Lyapunov candidate function

$$V(x) = x^T P x. \tag{34}$$

Differentiating along the closed-loop trajectories yields

$$\dot{V} = x^T \left[(\mathcal{A} - \mathcal{B}K)^T P + P(\mathcal{A} - \mathcal{B}K) \right] x. \tag{35}$$

Using the Riccati equation,

$$(\mathcal{A} - \mathcal{B}K)^T P + P(\mathcal{A} - \mathcal{B}K) = -(Q_N + K^T R K). \tag{36}$$

Therefore,

$$\dot{V} = -x^T (Q_N + K^T R K) x. \tag{37}$$

Since

$$Q_N \succeq 0$$

and

$$R > 0,$$

it follows that

$$\dot{V} \leq 0. \tag{38}$$

Consensus Convergence

Because

$$Q_N = (L \otimes I_n)^T Q (L \otimes I_n),$$

the condition

$$\dot{V} = 0$$

implies

$$(L \otimes I_n)x = 0. \tag{39}$$

Since the graph is connected,

$$x_1 = x_2 = \cdots = x_N = x^\star. \tag{40}$$

Hence all agents asymptotically reach consensus.

Nash Equilibrium Analysis

At equilibrium,

$$e_i = 0, \qquad \forall i. \tag{41}$$

Therefore,

$$u_i^\star = 0. \tag{42}$$

Suppose that agent i unilaterally deviates from the equilibrium strategy. Then

$$e_i \neq 0, \tag{43}$$

which implies

$$e_i^T Q e_i > 0. \tag{44}$$

Consequently,

$$J_i(u_i^\star, u_{-i}^\star) \leq J_i(u_i, u_{-i}^\star). \tag{45}$$

Thus no agent can reduce its own cost by changing its strategy alone. Therefore, the consensus configuration constitutes a Nash equilibrium.

Theorem 0.1. Consider the multi-agent system

$$\dot{x}_i = Ax_i + Bu_i + w_i$$

with distributed Nash controller

$$u_i^\star = -K \sum_{j \in \mathcal{N}_i} (x_i - x_j),$$

where

$$K = R^{-1}\mathcal{B}^T P$$

and P is the unique positive-definite solution of the Algebraic Riccati Equation. If the communication graph is connected and

$$Q > 0, \qquad R > 0,$$

then:

1. the closed-loop system is asymptotically stable,

2. all agents achieve consensus,

3. the consensus configuration is a Nash equilibrium,

4. the control law minimizes the global quadratic performance index.

Simulations

Figure 1 illustrates the evolution of the agents' position states. Although the agents start from different initial conditions, all trajectories gradually converge toward a common equilibrium value. This behavior confirms that the distributed Nash consensus controller successfully eliminates state disagreement among the agents. The smooth convergence without oscillatory behavior indicates an appropriately tuned LQR gain and stable closed-loop dynamics.

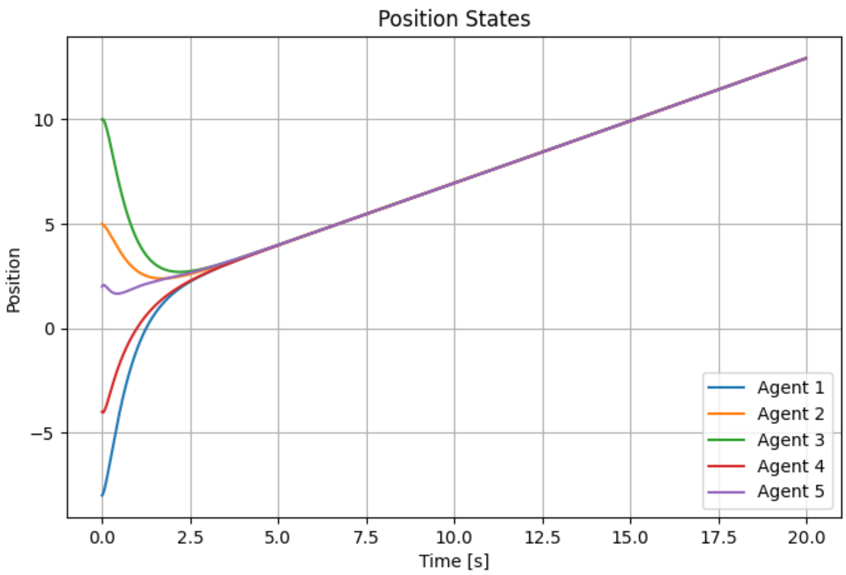

Figure 1: Θέσεις Πρακτόρων

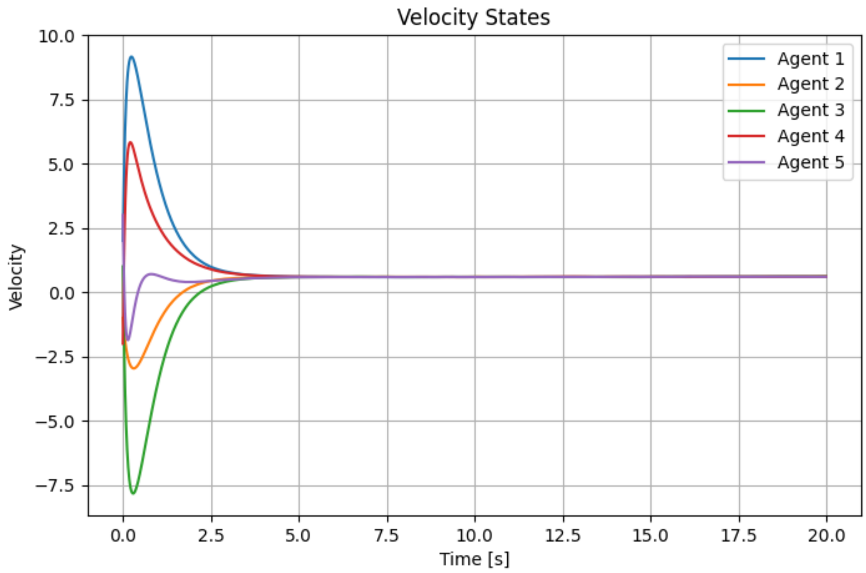

Figure 2: Velocities of every agent

The control inputs exhibit relatively large magnitudes during the transient phase, reflecting the effort required to reduce the initial disagreement among agents. As consensus is approached, the control actions progressively decrease and converge toward zero, indicating that no additional control effort is required once the equilibrium has been reached.

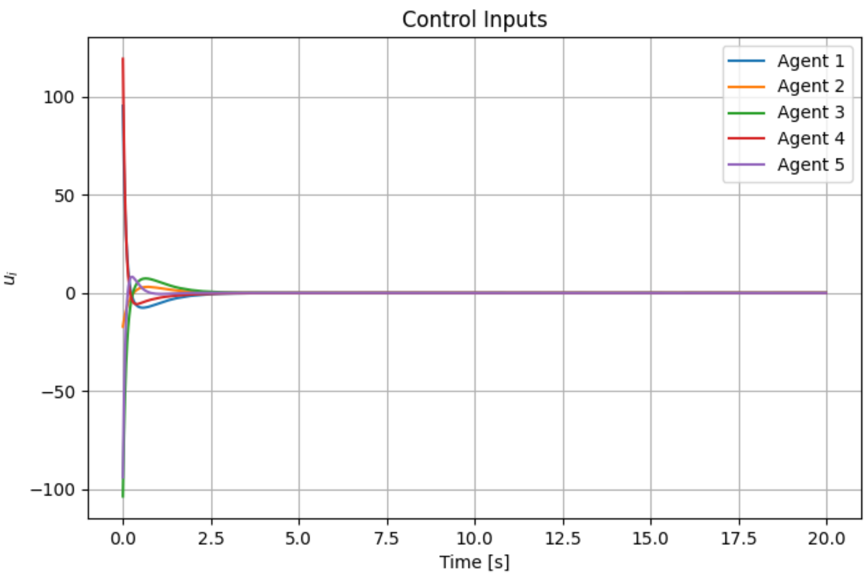

Figure 3: Inputs of every agent

The local performance indices decrease throughout the simulation, indicating that each agent successfully minimizes its individual cost function while interacting with neighboring agents. This behavior is consistent with the Nash equilibrium interpretation of the proposed framework.

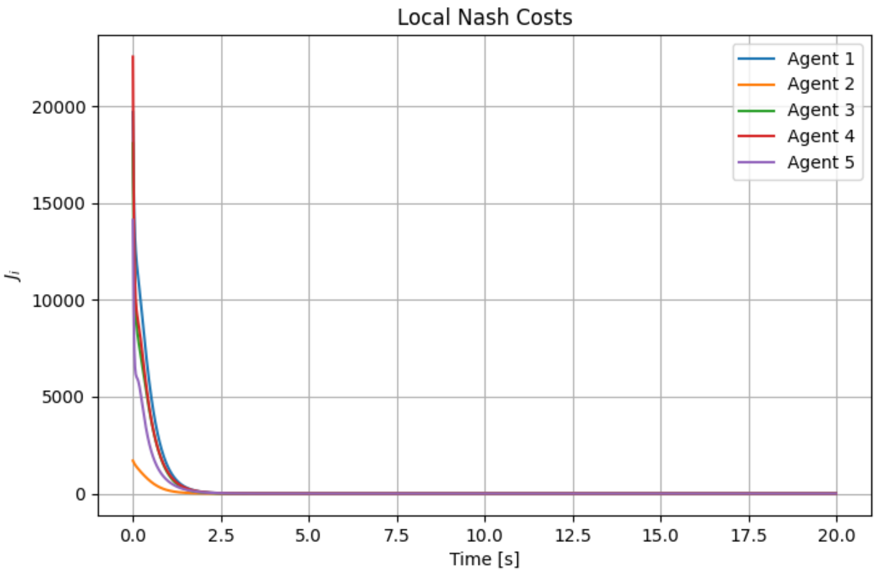

Figure 4: Cost functions of every agent

The consensus error trajectories decrease monotonically and asymptotically approach zero. This result demonstrates the effectiveness of the distributed feedback law in coordinating the agents and verifies the theoretical consensus condition derived in the stability analysis.

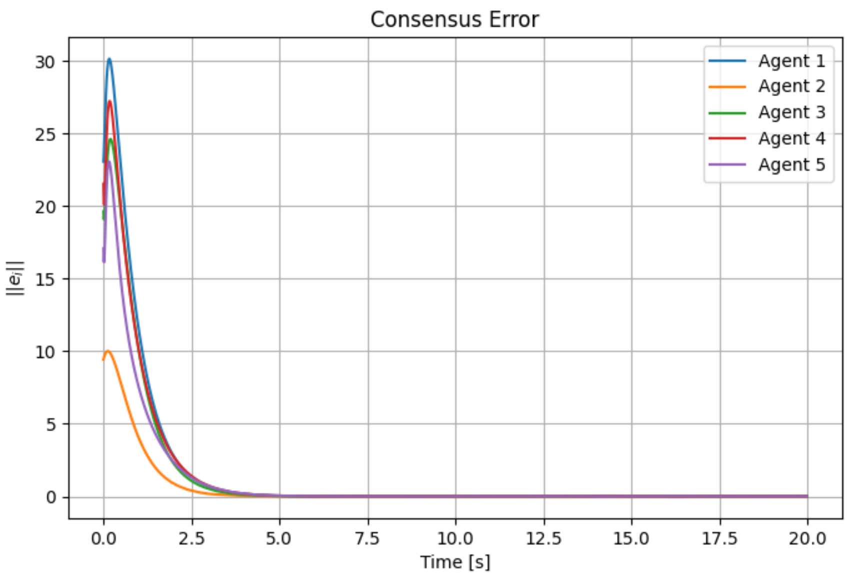

Figure 5: Consensus error of every agent

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